

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

Finding reasoning paths :

- 1. The anatomical snuff box is located on the lateral side of the hand , between the third and fourth metacarpal bones and the scaphoid .
- 2. The anatomical snuff box contains several important superficial vascular structures , including the radial and ulnar arteries .
- 3. The radial artery , responsible for supplying blood to the forearm , is a major artery in this area .
- 4. Damage to the ulnar artery , which is also superficial and visible in the anatomical snuff box , could lead to excessive bleeding .

Reasoning Process :

- 1. The anatomical snuff box is a bony depression on the lateral side of the hand , which serves as a landmark for identifying the arteries in this area .
- 2. The radial artery , which is superficial and runs along the posterolateral aspect of the wrist , is commonly associated with the anatomical snuff box .
- 3. The ulnar artery , which is deeper and runs along the posteromedial aspect of the forearm , is also found in this area .
- 4. In the context of a penetrating knife wound , the vessel that is most superficial and likely affected is the radial artery .
- 5. Given the location and the nature of the injury , the radial artery , being superficial and in close proximity to the skin surface , is the most likely to be damaged , leading to excessive bleeding .

Conclusion :

The radial artery is most likely responsible for the excessive bleeding in this case , as it is the superficial artery in the anatomical snuff box and is commonly affected in this type of injury .

</think>

< answer >

Radial artery

</answer>

